

Cover Letter

Candidate: Ahmet Halit Ünsal

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Position: Uçuş Kontrol ve Model Tabanlı Tasarım Lider Mühendisi

Target Organization: TUSAŞ (Türk Havacılık Uzay Sanayii) (Ankara (Kahramankazan), Turkey)

Job URL: <https://vizyonergenc.com/ilan/tusas-ucus-kontrol-mbd-301>

Dear Hiring Team & Leadership at TUSAŞ (Türk Havacılık Uzay Sanayii),

I am writing to express my strong interest in the Uçuş Kontrol ve Model Tabanlı Tasarım Lider Mühendisi position at TUSAŞ (Türk Havacılık Uzay Sanayii).

With over 15 years of professional engineering experience - including 8+ years leading and directing multi-disciplinary engineering organizations of up to 30 engineers across 4 team leads - my background directly aligns with TUSAŞ (Türk Havacılık Uzay Sanayii)'s requirements for high-reliability embedded software architecture, Model-Based Design (MBD), and mission-critical powertrain controls.

Throughout my leadership tenure across automotive and defense electronics programs (NISO Technology, ECEMTAG, TÜBİTAK, TÜMOSAN), I have directed full product lifecycles from initial concept and dynamic modeling to dyno characterization and mass production. I maintain strict enforcement of ISO 26262 ASIL D functional safety, ASPICE Level 2/3 processes, UN R155/R156 cybersecurity (CSMS), and AUTOSAR layered architectures across FreeRTOS and bare-metal platforms.

On the technical front, I bring hands-on mastery in MATLAB/Simulink/Stateflow physical modeling, PMSM/IPMSM Field-Oriented Control (FOC, MTPA, flux-weakening), and complex EV powertrain controls (VCU, MCU, Traction Inverters, OBC, BMS). Complementing this is extensive experience establishing automated Hardware-in-the-Loop (HIL) simulation testbenches (dSpace MABX, Lauterbach Trace32) and Git-based CI/CD pipelines that dramatically compress verification cycles and eliminate integration bottlenecks.

Beyond embedded systems, my end-to-end technical leadership is demonstrated in platforms like EduTrace (edutrace.net), highlighting my ability to scale complex architectures and bridge software, hardware, and product teams. I would welcome the opportunity to discuss how my technical ownership, executive leadership, and delivery focus can accelerate the engineering milestones at TUSAŞ (Türk Havacılık Uzay Sanayii).

Sincerely,

Ahmet Halit Ünsal

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